

Tyrone Marhguy

Computer Engineering — Computer Architecture / RTL / FPGA / ASIC / Verification

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SUMMARY

Computer Engineering junior building from transistor-level CMOS and physical design through RTL, FPGA systems, CPU/GPU microarchitecture, networking, formal verification, and ASIC implementation. Built custom processors, a programmable 3D GPU, low-latency datapaths, SRAM/MAC blocks, and production EDA fixes.

EDUCATION

University of Pennsylvania

B.S.E. Computer Engineering

Philadelphia, PA

Aug. 2024 – May 2028

- Coursework: Computer Organization & Design, VLSI, Digital Logic, Embedded Systems

EXPERIENCE

Hardware & Firmware Engineer

Vero Electric

May 2026 – Aug. 2026

Philadelphia, PA

- Developed embedded C/C++ fault-detection firmware and a **BQ79616 16-channel battery-monitoring front end** for a **240 kWh / 120 kW** system; closed PCB ERC with **0 errors**; bring-up with lab instrumentation.

Open-Source EDA Contributor

OpenROAD, Verilator, LibreLane, OpenFPGA

2026 – Present

C++ / Python / EDA Infrastructure

- Merged fixes into production EDA tools (**OpenROAD** parser, **Verilator** peak-memory accounting, LibreLane/Yosys reporting); shipped in releases **3.0.8** and **3.0.10**.

Software Engineer Intern

Aragorn AI

May 2026 – Aug. 2026

Remote

- Shipped production backend services and REST APIs; resolved cross-service failures and hardened system boundaries.

SELECTED HARDWARE PROJECTS

- 1. Tomato — Custom 32-bit Polymorphic CPU** | SystemVerilog / FPGA / KiCad / Formal Website | GitHub
 - Architected a custom 32-bit CPU around two programmable **LUT3 planes + 8-way carry/flag select (524,288 ALU configurations)**, three-source datapath, and microcoded control; working FPGA computer with assembler and TomatoOS over HDMI.
 - **130B Verilator vectors** + formal 1/8/32-bit proofs; fabricated discrete ALU PCB; full 74xx/discrete machine toward bring-up.
- 2. Out-of-Order RISC-V CPU (RV64IMAC)** | SystemVerilog / RISC-V / AXI4-Lite GitHub
 - RV64IMAC (riscv64xO3): **2-wide fetch/decode/issue/commit**, gshare + BTB + RAS, AXI4-Lite; Verilator/cocotb/riscv-tests with Yosys/OpenLane2 feasibility.
- 3. Pineapple GPU P1 — Standalone Programmable 3D GPU** | Verilog / Artix-7 / DVI GitHub
 - Vertex/fragment shading, rasterization, depth, texture, double-buffered **320x180**; interactive pineapple/cube scenes on a Nexys A7 via FOSS Yosys/nextpnr.
- 4. Low-Latency FPGA Networking & NASDAQ ITCH** | SystemVerilog / RMII / cocotb UDP/IP | ITCH
 - Built a **100 Mbps RMII/MAC/IPv4/UDP** pipeline and RTL ITCH 5.0 parser with 512-entry BRAM book; closed at **100 MHz**; **10M-message** regression, zero BBO mismatches vs Python/cocotb model.
- 5. ASIC / Full-Custom Datapaths** | Sky130 / 22 nm HP / SPICE MAC | SRAM
 - Designed a BF16 multiply-to-FP32-accumulate MAC with LibreLane/OpenLane hardening and a transistor-level **16x4 6T SRAM** with StrongARM sensing; NGSpice-swept simulated readback to **4.571 GHz**.

TECHNICAL SKILLS

Architecture / RTL: SystemVerilog/Verilog, VHDL, CPU/GPU/SoC microarchitecture, pipelining, superscalar/OoO design, ISA/datapath design, branch prediction, memory systems, CDC

Verification / ASIC: UVM, SVA, cocotb, formal, Verilator; Vivado, Yosys, OpenROAD, LibreLane/OpenLane, Sky130, synthesis, STA, P&R, DRC/LVS, SPICE, KiCad, Electric VLSI

Programming / Interfaces: Python, C/C++, Tcl, Bash, Linux; Ethernet/RMII/IPv4/UDP, AXI4-Lite, UART, SPI, I2C